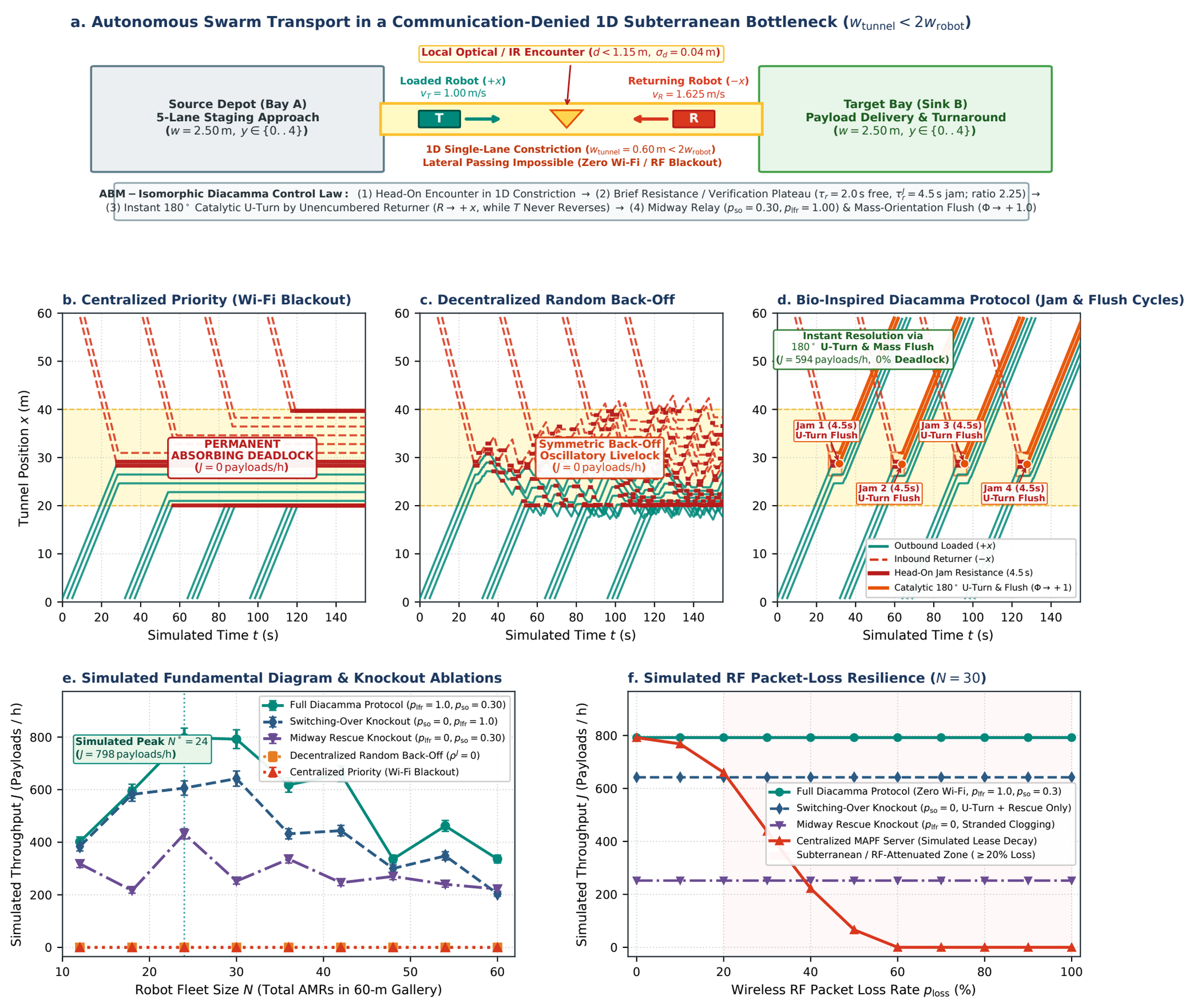

a. Autonomous Swarm Transport in a Communication-Denied 1D Subterranean Bottleneck ($w_{tunnel} < 2w_{robot}$)
Local Optical / IR Encounter ($d < 1.15$ m, $\sigma_d = 0.04$ m)
Source Depot (Bay A)
5-Lane Staging Approach
($w = 2.50$ m, $y \in \{0..4\}$)
Loaded Robot ($+x$)
$v_T = 1.00$ m/s
Returning Robot ($-x$)
$v_R = 1.625$ m/s
T
R
Target Bay (Sink B)
Payload Delivery & Turnaround
($w = 2.50$ m, $y \in \{0..4\}$)
1D Single-Lane Constriction ($w_{tunnel} = 0.60$ m $< 2w_{robot}$)
Lateral Passing Impossible (Zero Wi-Fi / RF Blackout)
ABM – Isomorphic Diacamma Control Law : (1) Head-On Encounter in 1D Constriction → (2) Brief Resistance / Verification Plateau ($\tau_r = 2.0$ s free, $\tau_r' = 4.5$ s jam; ratio 2.25) → (3) Instant 180° Catalytic U-Turn by Unencumbered Returner ($R \to +x$, while T Never Reverses) → (4) Midway Relay ($p_{so} = 0.30$, $p_{lfr} = 1.00$) & Mass-Orientation Flush ($\Phi \to +1.0$)
b. Centralized Priority (Wi-Fi Blackout)
PERMANENT
ABSORBING DEADLOCK
($J = 0$ payloads/h)
Tunnel Position x (m)
Simulated Time t (s)
c. Decentralized Random Back-Off
Symmetric Back-Off
Oscillatory Livelock
($J = 0$ payloads/h)
Simulated Time t (s)
d. Bio-Inspired Diacamma Protocol (Jam & Flush Cycles)
Instant Resolution via
180° U-Turn & Mass Flush
($J = 594$ payloads/h, 0% Deadlock)
Jam 1 (4.5s)
U-Turn Flush
Jam 2 (4.5s)
U-Turn Flush
Jam 3 (4.5s)
U-Turn Flush
Jam 4 (4.5s)
U-Turn Flush
Outbound Loaded ($+x$)
Inbound Returner ($-x$)
Head-On Jam Resistance (4.5 s)
Catalytic 180° U-Turn & Flush ($\Phi \to +1$)
Simulated Time t (s)
e. Simulated Fundamental Diagram & Knockout Ablations
Simulated Peak $N^* = 24$
($J = 798$ payloads/h)
Full Diacamma Protocol ($p_{lfr} = 1.0$, $p_{so} = 0.30$)
Switching-Over Knockout ($p_{so} = 0$, $p_{lfr} = 1.0$)
Midway Rescue Knockout ($p_{lfr} = 0$, $p_{so} = 0.30$)
Decentralized Random Back-Off ($\rho^J = 0$)
Centralized Priority (Wi-Fi Blackout)
Simulated Throughput J (Payloads / h)
Robot Fleet Size N (Total AMRs in 60-m Gallery)
f. Simulated RF Packet-Loss Resilience ($N = 30$)
Full Diacamma Protocol (Zero Wi-Fi, $p_{lfr} = 1.0$, $p_{so} = 0.3$)
Switching-Over Knockout ($p_{so} = 0$, U-Turn + Rescue Only)
Midway Rescue Knockout ($p_{lfr} = 0$, Stranded Clogging)
Centralized MAPF Server (Simulated Lease Decay)
Subterranean / RF-Attenuated Zone (≥ 20% Loss)
Simulated Throughput J (Payloads / h)
Wireless RF Packet Loss Rate p_{loss} (%)